

End-of-Module Exams 2025 Fall Semester

Signature: _____

I hereby confirm that all the pages of my exam paper
have been stapled to this cover sheet.

Number of supplementary answer sheets attached:

Total number of added pages:

Module: w.BA.XX.3DA-WIN.XX / Data Analytics

Exam date / time: 22.01.2026 / 10:00 - 11:00

Exam duration: 60 Minutes

Total number of pages: 24 (Incl. cover sheet)

Resources allowed: Open book

Calculator: Approved calculator according to 'Guidelines on Supplementary Materials'

Dictionary: Yes

General information:

- Use a black or blue pen (indelible ink) to write your answers.
- Write your name on every page.
- Only write on the front side of each page.
- Supplementary answer sheets are provided by the invigilators. These must be clearly numbered and the number of supplementary answer sheets used must be noted on the cover sheet. Do not use your own sheets of paper.
- Sign the cover sheet before you hand in the exam.
- If you encounter an exam question that does not seem clear to you, you may feel the need to make assumptions. In such a case, summarize your assumptions in your answer.
- For remarks pertaining to specific questions, please refer to information included in the task.
- Visits to the restroom (toilet) are only permitted if an exam lasts 90 minutes or more.
- Invigilators are not allowed to respond to any questions related to the content of the exam.
- If you wish to hand in your exam early, you may do so until 20 minutes before the official end of the examination.

Marked by:	Total points:	Grade:

End-of-Module Exams: 2025 Fall Semester
Module Code: w.BA.XX.3DA-WIN.XX

Zürich University
of Applied Sciences



**School of
Management and Law**

Last Name:

First Name:

Answer Sheet – Please note that only the answers on this sheet will be assessed.

Grading Table

Performance Assessment	Weighting	Grade	Weighted Grade
Performance Assessment 1 Written Exam	70%		
Performance Assessment 2 Project Work (Group Work)	30%		
Performance Assessment 3 Submission of the solutions of at least 10 weekly exercises	Pass/Fail	Pass/Fail	-
		(Grade on cover sheet)	
Module grade (Evento)			↔ Rounded ↔

Marking Table

No.	Task	Points	Corrector's, initials
1	Single-Choice Questions		
Total Points			
Grade			

Overview

Task	Description	Points
1	Single-Choice Questions	60

TOTAL: 60 points

Last Name:

First Name:

Answer-Sheet (please mark the correct answer per question):

Question	a)	b)	c)	d)
Q1	☐	☐	☐	☐
Q2	☐	☐	☐	☐
Q3	☐	☐	☐	☐
Q4	☐	☐	☐	☐
Q5	☐	☐	☐	☐
Q6	☐	☐	☐	☐
Q7	☐	☐	☐	☐
Q8	☐	☐	☐	☐
Q9	☐	☐	☐	☐
Q10	☐	☐	☐	☐
Q11	☐	☐	☐	☐
Q12	☐	☐	☐	☐
Q13	☐	☐	☐	☐
Q14	☐	☐	☐	☐
Q15	☐	☐	☐	☐

Question	a)	b)	c)	d)
Q16	☐	☐	☐	☐
Q17	☐	☐	☐	☐
Q18	☐	☐	☐	☐
Q19	☐	☐	☐	☐
Q20	☐	☐	☐	☐
Q21	☐	☐	☐	☐
Q22	☐	☐	☐	☐
Q23	☐	☐	☐	☐
Q24	☐	☐	☐	☐
Q25	☐	☐	☐	☐
Q26	☐	☐	☐	☐
Q27	☐	☐	☐	☐
Q28	☐	☐	☐	☐
Q29	☐	☐	☐	☐
Q30	☐	☐	☐	☐

Question Section – Please note that this question paper will not be assessed.

General Information

- This exam consists of **single-answer questions only**.
- This means there is only one correct answer for each question.
- The possible answers for each question are listed as (a), (b), (c), and (d).
- If you provide more than one answer for a question, you will receive no marks for that question.
- If you do not answer a question, you will receive no marks for that question.
- Use the answer sheet for your solutions (**answer section** of the exam). This page belongs to the **question section** of the exam.

Overview

Task	Description	Points
1	Single-Answer Questions	60

TOTAL: 60 points

Single-Answer Questions

(60 points)

Categorization of Data & Data Collection Methods

Q1: Consider the following table containing bicycle data. Which of the statements with regard to the “scale of measure” of these variables is **not** correct?

2 points

Brand	Model	Frame Size	Weight kg	Gears	Type	Brake System	Price
Trek	Dornane SL6	56	8.9	22	Road Bike	Disc	3299
Giant	TCR Advanced 1	54	8.1	22	Road Bike	Rim	2899
Canyon	Endurace CF 7	58	8.4	22	Endurance Bike	Disc	2499
Specialized	Roubaix Sport	56	8.8	22	Road Bike	Disc	3100
Scott	Addict 30	52	8.5	20	Road Bike	Rim	2699
Trek	Checkpoint ALR 5	54	9.4	22	Gravel Bike	Disc	2599

- a) The variable *Brand* has unordered category labels and is, therefore, nominal scaled.
- b) The variable *Price* is numeric with a meaningful zero and is, therefore, ratio scaled.
- c) The variable *Type* has ordered category labels and is, therefore, ordinal scaled.
- d) The variable *Model* has unordered category labels and is, therefore, nominal scaled.

Q2: The text item "1965" (i.e., the year of manufacture of the classic motorbike Triumph Bonneville T120) must be extracted from the following HTML file using an XPath. Which is the correct XPath for extracting this value?

2 points

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <title>Classic Motorbikes</title>
</head>
<body>
  <table border="1">
    <tr>
      <th>Make</th>
      <th>Model</th>
      <th>Type</th>
      <th>Engine</th>
      <th>Country</th>
      <th>Year</th>
    </tr>
    <tr>
      <td>Honda</td>
      <td>CB450</td>
      <td>Standard</td>
      <td>444 cc</td>
      <td>Japan</td>
      <td>1969</td>
    </tr>
    <tr>
      <td>Triumph</td>
      <td>Bonneville T120</td>
      <td>Classic</td>
      <td>649 cc</td>
      <td>UK</td>
      <td>1965</td> <!-- Target cell for XPath -->
    </tr>
  </table>
</body>
</html>
```

- a) `html/head/table/tr[3]/td[5]/text()`
- b) `html/body/table/tr[3]/td[6]/text()`
- c) `html/body/table/tr[2]/td[6]/text()`
- d) `html/head/table/tr[2]/td[5]/text()`

Q3: Which statement about the following Python code is not correct?

2 points

```
import json
import requests
from pandas import json_normalize

# Step 1
url = 'http://overpass-api.de/api/interpreter'
query = f"""
[out:json];
area["ISO3166-1"~"CH"][[admin_level=2]];
node["shop"~"hairdresser"]({area});
out;"""

# Step 2
r = requests.get(url, params={'data': query})
data = r.json()['elements']

# Step 3
with open('query_result.json', 'w') as json_file:
    json.dump(data, json_file)

# Step 4
df = json_normalize(data)

# Step 5
df.head()
```

- a) The query requests all nodes tagged as "hairdresser" within Switzerland using ISO country code "CH".
- b) The retrieved data from the Overpass API is stored in JSON format and saved as a file named "query_result.json".
- c) The json_normalize() function converts the JSON data into a pandas DataFrame.
- d) The code sends data to the Overpass API to create new "hairdresser" entries in Switzerland.

Data Types, Data Structures, & File Formats in Python

Q4: Which data types are shown by the print() function in the code below?

2 points

```
a = 5 + 3.0
b = '12' + str(8)
c = len(['Yamaha', 'BSA', 'Norton'])
d = (10 > 3) and isinstance(4.5, float)
e = len('Simson') > 5
f = type(b) == str
g = (10 > 4) or (5 < 10)

print(type(a), type(b), type(c), type(d), type(e), type(f), type(g))
```

- a) float, str, int, bool, bool, bool, bool
- b) int, str, float, bool, int, bool, bool
- c) float, int, int, bool, bool, str, bool
- d) float, str, int, bool, bool, str, int

Q5: What is the output of the following Python code? Note that the `list()` function simply converts an object to a list, for example, `list({1,2,3,4,5})` returns `[1, 2, 3, 4, 5]`.

2 points

```
obj = {'key': [1, 2, 3, 4],  
      'name': ['Paul', 'Lotta', 'Anne', 'Joe'],  
      'gender': ['M', 'F', 'F', 'M'],  
      'age': [21, 27, 33, 31],  
      'city': ['Berlin', 'Porto', 'Paris', 'Madrid']}  
  
print(list(obj.keys()))
```

- a) `[[1, 2, 3, 4],
 ['Paul', 'Lotta', 'Anne', 'Joe'],
 ['M', 'F', 'F', 'M'],
 [21, 27, 33, 31],
 ['Berlin', 'Porto', 'Paris', 'Madrid']]`
- b) `[1, 2, 3, 4]`
- c) `['key', 'name', 'gender', 'age', 'city']`
- d) `[0,1,2,3,4]`

Q6: Consider the following list (named `obj_01`) and set (named `obj_02`). What is the output of the `print()` function?

2 points

```
obj_01 = ['Beagle', 1234, 'Bulldog', 'Beagle', 1234]  
obj_02 = set(obj_01)  
print(obj_01, obj_02)
```

- a) `['Beagle', 1234, 'Bulldog', 'Beagle', 1234] {1234, 'Bulldog', 'Beagle'}`
- b) `['Beagle', 1234, 'Bulldog'] {'Beagle', 1234, 'Bulldog'}`
- c) `['Beagle', 1234, 'Bulldog', 'Beagle', 1234] {'Beagle', 1234, 'Bulldog', 'Beagle', 1234}`
- d) `['Beagle', 1234, 'Bulldog'] {'Beagle', 1234, 'Bulldog', 'Beagle', 1234}`

Q7: What type will the following expression return?

2 points

```
obj = {  
    'pets': ['cat', 'dog', 'bird'],  
    'numbers': (1, 2, 3, 4),  
    'colors': {'red', 'green', 'blue'},  
    'info': {'name': 'Max', 'age': 5}  
}  
  
print(type(list(obj.values())[3]))
```

- a) `<class 'list'>`
- b) `<class 'dict'>`
- c) `<class 'set'>`
- d) `<class 'tuple'>`

Q8: What type will the following expression return?

2 points

```
import pandas as pd
import json

# Step 1: Create data
data = {
    "employees": [
        {"name": "Alice", "age": 30, "city": "Zurich"},
        {"name": "Bob", "age": 25, "city": "Bern"},
        {"name": "Charlie", "age": 35, "city": "Basel"}
    ]
}

# Step 2: Write data to a file
with open("employees.json", "w") as f:
    json.dump(data, f, indent=4)

# Step 3: Read the file
df = pd.read_json("employees.json")

# Step 4: Check the type of the object 'df'
print(type(df))
```

- a) <class 'json'>
- b) <class 'pandas.core.series.Series'>
- c) <class 'dict'>
- d) <class 'pandas.core.frame.DataFrame'>

Data Preparation & Combining & Organizing Data

Q9: Consider the pandas DataFrame below (name = data) and the Python code for the selection. Which tractor brands are selected? Remember that the | operator inside the .loc method means 'OR'.

2 points

DataFrame 'data'

	id	brand	country	horsepower	category	price (k€)
0	1	John Deere	USA	180	A	95
1	2	Fendt	Germany	240	B	120
2	3	Massey Ferguson	UK	210	C	110
3	4	New Holland	Italy	160	B	88
4	5	Claas	NaN	220	A	115
5	6	Fendt	Germany	240	B	120
6	7	Kubota	NaN	130	A	75

```
# Selection condition
subset = data.dropna().drop_duplicates().loc[
    (data.horsepower >= 200) | (data.country == 'Germany')
]

print(list(subset.brand))
```

- a) ['Fendt', 'Massey Ferguson', 'Fendt']
- b) ['Fendt', 'Massey Ferguson', 'Claas', 'New Holland']
- c) ['Fendt', 'Massey Ferguson', 'Claas']
- d) ['Fendt', 'Massey Ferguson', 'Claas', 'Kubota']

Q10: Consider the following Python code for creating and combining two data frames named df1 and df2. How many rows does the output data frame df3 contain, and how many columns? Note that “number of columns” means without the index (i.e., 0, 1, 2, ...).

2 points

```
import pandas as pd

# 1st DataFrame
df1 = pd.DataFrame({
    'Brand': ['John Deere', 'Fendt', 'Massey Ferguson', 'Claas'],
    'Country': ['USA', 'Germany', 'UK', 'Germany'],
    'Horsepower': [180, 240, 210, 220]
})

# 2nd DataFrame
df2 = pd.DataFrame({
    'Brand': ['New Holland', 'Kubota', 'Case IH', 'Valtra'],
    'Country': ['Italy', 'Japan', 'USA', 'Finland'],
    'Horsepower': [160, 130, 250, 200]
})

# Combining the DataFrames
df3 = pd.concat([df1, df2], ignore_index=True)

print(df3.shape)
```

- a) The data frame df3 has 7 rows and 5 columns.
- b) The data frame df3 has 5 rows and 5 columns.
- c) The data frame df3 has 8 rows and 3 columns.
- d) The data frame df3 has 4 rows and 6 columns.

Q11: Consider the pandas DataFrame below (named df) showing different restaurant reviews, and the Python code. When running the Python code, how many restaurants are marked with a value of 1 (meaning "recommended") in the new variable? 2 points

DataFrame 'df'

	id	review text	city	avg price
0	1	EXCELLENT FOOD, QUICK SERVICE.	Zurich	45
1	2	FRIENDLY STAFF, BUT THE PLACE WAS CROWDED.	Bern	38
2	3	THE BURGER WAS TOO SALTY AND COLD.	Basel	32
3	4	CLEAN TABLES, GREAT COFFEE.	Zurich	40
4	5	DIRTY DISHES, I WON'T COME BACK HERE.	Bern	35
5	6	VERY NICE PLACE WITH A LARGE MENU.	Zurich	42
6	7	FRIENDLY WAITER, DOGS ARE ALLOWED.	Basel	37
7	8	BEAUTIFUL TERRACE WITH A LAKE VIEW.	Zurich	60

```
# Create new variable 'recommended'
df['review_text'] = df['review_text'].str.upper()
pattern = '(FRIENDLY)|(CLEAN)|(BEAUTIFUL)|(EXCELLENT)'
df['recommended'] = df['review_text'].str.contains(pat=pattern).astype(int)
print(df['recommended'].sum())
```

- a) 3
- b) 5
- c) 4
- d) 6

Q12: Consider the following data frames named 'df1' and 'df2' and the Python code for the combination of these data frames. How many rows does the output data frame 'df3' contain and how many columns? Note, that "number of columns" means without the index (i.e., 0,1,2,3,4). 2 points

dataframe 'df1'

	brand	horsepower	country
0	John Deere	180	USA
1	Fendt	240	Germany
2	Massey Ferguson	210	UK
3	Claas	220	Germany
4	Kubota	130	Japan

dataframe 'df2'

	brand	category	price (k€)
0	John Deere	A	95
1	Fendt	A	120
2	Claas	B	110
3	New Holland	B	105
4	Valtra	A	100

```
df3 = df1.merge(df2[['brand', 'price (k€)']], on='brand')
print(df3)
```

- a) The data frame df3 has 4 rows and 5 columns.
- b) The data frame df3 has 5 rows and 5 columns.
- c) The data frame df3 has 4 rows and 6 columns.
- d) The data frame df3 has 3 rows and 4 columns.

Exploratory Data Analysis (EDA)

Q13: Consider the following contingency table. It contains data from a survey in which respondents (men and women) stated whether they use public transport or not. Which statement with regard to this contingency table is **not** correct?
Note: The numbers shown in the answer options are rounded to one decimal place.

2 points

	Men	Women	Sum
I use public transport	72	88	160
I do not use public transport	28	42	70
Sum	100	130	230

- a) 69.6% of all respondents use public transport.
- b) 18.3% of all respondents are women who do not use public transport.
- c) 31.3% of all respondents are men who use public transport.
- d) 32.4% of all respondents do not use public transport.

Q14: Consider the following data representing a sample of smartphone prices (in CHF). Which is the correct value of the sample standard deviation of the prices?
Note: The values in the answer options are rounded to one decimal place.

2 points

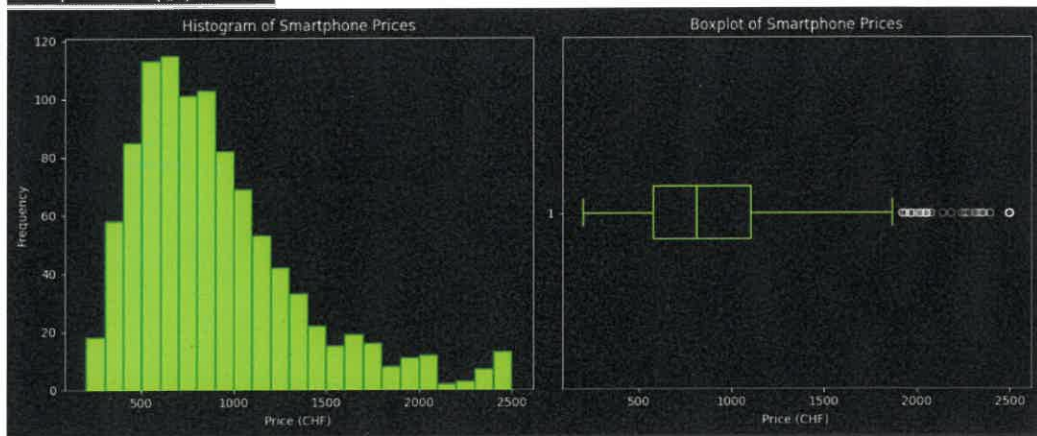
	Brand	Model	Price (CHF)
0	Apple	iPhone 15	1099
1	Samsung	Galaxy S23	999
2	Google	Pixel 8	899
3	OnePlus	11 Pro	799
4	Xiaomi	13T	699

- a) 172.5
- b) 158.1
- c) 164.3
- d) 170.9

Q15: Consider the following graphics showing the distribution of smartphone prices (in CHF) and the measures of central tendency of the price distribution. Which statement with regard to these graphics and measures is not correct?

2 points

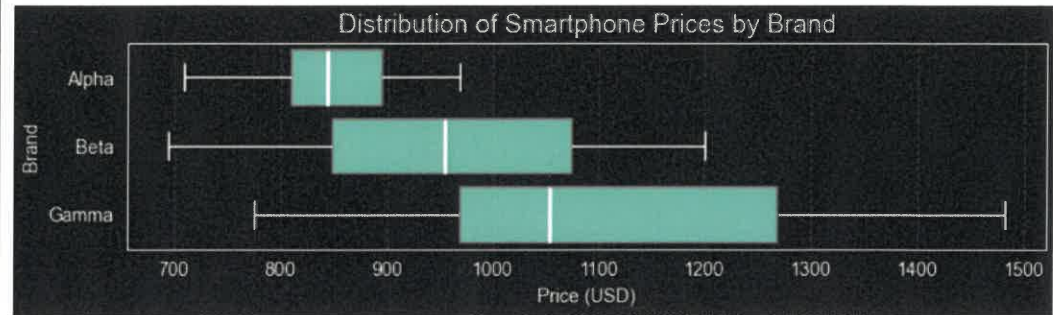
1st quantile (Q1): 579
Median: 810
3rd quantile (Q3): 1106



- a) The interquartile range (IQR) of smartphone prices is CHF 527.
- b) It can be concluded from the graphs and data that the mean smartphone price is < CHF 810.
- c) The calculation CHF 1106 minus CHF 579 provides the spread of the middle 50% of smartphone prices, which is also shown as the box in the boxplot.
- d) The smartphone prices have a right-skewed (positively skewed) distribution.

Q16: The following grouped boxplot shows the distribution of smartphone prices (in USD) for three brands: Alpha, Beta, and Gamma. Which of the following statements are correct?

2 points



Statements:

- (1) Gamma shows a larger price variation (indicated by the IQR) compared to Alpha and Beta.
- (2) Gamma's price distribution is right-skewed.
- (3) The average price of Gamma (although not shown in the graphic) must be higher than its median price.
- (4) Alpha's price distribution is relatively compact, indicating lower price variation across its models.

- a) Only statements (1), (2), and (3) are correct.
- b) Only statements (1), (3), and (4) are correct.
- c) Only statements (1), (2), and (4) are correct.
- d) All four statements (1), (2), (3), (4) are correct.

Representation & Analysis of Geographical Data

Q17: Which statement with regard to the GIS data format below is not correct?

2 points

Point		POINT (30 10)
LineString		LINESTRING (30 10, 10 30, 40 40)

- a) This data format is used to store vector data.
- b) The numbers, e.g., 30 10, 10 30, 40 40, refer to the spatial coordinates of points.
- c) This GIS data format is referred to as GeoJSON.
- d) This data format can also be used to store point and line geometry types.

Q18: Consider the following Python code and the parts of the code denoted Part (1) ... Part (4). Which statement below with regard to this Python code is not correct?

2 points

```
import json
import urllib
import requests
import pandas as pd

# Part (1)
base_url = "https://api3.geo.admin.ch/rest/services/api/SearchServer?"

# Part (2)
parameters = {
    "searchText": "Bahnhofstrasse 10, 8001 Zürich",
    "origins": "address",
    "type": "locations",
}

# Part (3)
r = requests.get(f"{base_url}{urllib.parse.urlencode(parameters)}")

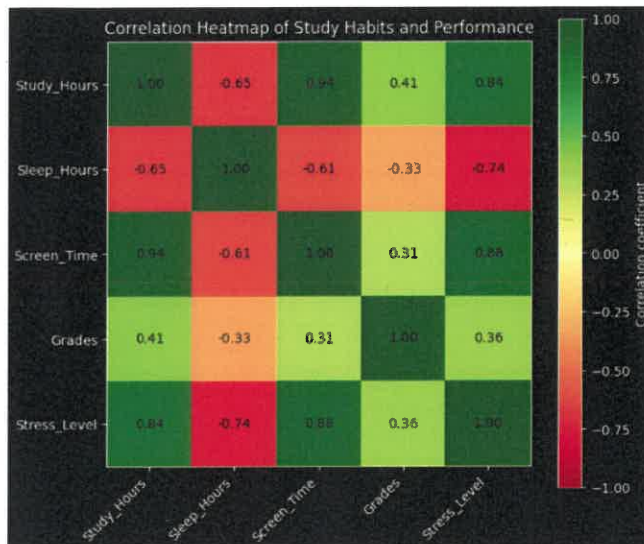
# Part (4)
data = json.loads(r.content)
df = pd.DataFrame.from_dict(list(data.values())[0][0], orient='columns')
df.iloc[[1, 4, 5, 6, 11, 12], :1]
```

- a) The Python code is used to geocode a single address using the GeoAdmin Web API.
- b) The requests.get() method in Part (3) sends a GET request to the server via the constructed URL.
- c) In Part (4), df.iloc[[1,4,5,6,11,12],:1] is used to select relevant data from the server response stored in the data frame named 'df'.
- d) The focus of the GeoAdmin Web API is to offer geospatial data, maps, and services for different European countries.

Statistical Hypothesis Testing

Q19: With reference to the correlation heatmap shown below, which statement about students' study habits and performance is **not** correct? Note: Although not shown in the plot, all correlations are statistically significant at the 5% significance level.

2 points



- a) Students who study more tend to have slightly less sleep, suggesting a moderate negative correlation between Study_Hours and Sleep_Hours.
- b) The correlation between Study_Hours and Grades is close to zero, implying that studying more does not strongly affect grades.
- c) Screen_Time and Stress_Level show a positive correlation, meaning higher screen usage is linked to greater stress.
- d) Sleep_Hours and Stress_Level are negatively correlated, indicating that less sleep is generally associated with higher stress.

Q20: Consider the previously shown contingency table of observed frequencies about people's preferences regarding public transport. For a Chi-squared test, the expected frequencies must be calculated. Which of the following sets of expected frequencies is correct? Note: All values are rounded to two decimal places.

2 points

	Men	Women	Sum
I use public transport	72	88	160
I do not use public transport	28	42	70
Sum	100	130	230

a)

Expected Frequencies:

	Men	Women
I use public transport	69.57	90.43
I do not use public transport	30.43	39.57

b)

Expected Frequencies:

	Men	Women
I use public transport	62.57	96.43
I do not use public transport	34.43	38.57

c)

Expected Frequencies:

	Men	Women
I use public transport	61.57	95.43
I do not use public transport	33.43	37.57

d)

Expected Frequencies:

	Men	Women
I use public transport	63.57	99.43
I do not use public transport	36.43	31.57

Regression Analysis	
Q21: Given the regression equation for smartphone prices based on storage capacity and camera resolution shown below, what does this equation imply? $y = 3.5x_1 + 25x_2 + 200$ where: • y = estimated smartphone price (in \$) • x_1 = storage capacity (in GB) • x_2 = camera resolution (in megapixels) a) For every increase of 1 GB in storage, the smartphone price increases by \$25, holding camera resolution constant. b) For every increase of 1 megapixel in camera resolution, the smartphone price increases by \$3.50, holding storage constant. c) For every increase of 1 GB in storage, the smartphone price increases by \$3.50, holding camera resolution constant. d) For every increase of 1 megapixel in camera resolution, the smartphone price increases by \$200, holding storage constant.	2 points
Q22: Imagine you have two numerical variables. You perform a correlation analysis and a regression analysis based on these two variables. If the coefficient of determination (R^2) equals 1, then the correlation coefficient (r) ... a) ... must also be equal to 1. b) ... can be either -1 or +1. c) ... can be any value between -1 and +1. d) ... must be -1.	2 points

Q23: Consider the multiple linear regression model below, which estimates smartphone prices based on three technical specifications: storage capacity, camera resolution, and battery capacity. Based on this regression equation, what is the estimated price of a smartphone with 128 GB storage, 48 MP camera resolution, and a 4000 mAh battery? Note that the values in the answer options are rounded to two decimal places. $y = 120 + 2.75x_1 + 8.60x_2 + 0.15x_3$ where: • y = estimated smartphone price (in USD) • x_1 = storage capacity (in GB) • x_2 = camera resolution (in megapixels) • x_3 = battery capacity (in mAh) a) 1250.50 b) 1385.75 c) 1484.80 d) 1520.25	2 points									
Classification Analysis										
Q24: Consider the following confusion matrix of a classification model in which TN = True Negative, FN = False Negative, FP = False Positive, and TP = True Positive. Which is the correct value of the "Recall" (rounded to four decimal places)? ----- Confusion Matrix: <table><tr><td></td><td>Predicted Positive</td><td>Predicted Negative</td></tr><tr><td>Actual Positive</td><td>TP = 950</td><td>FN = 120</td></tr><tr><td>Actual Negative</td><td>FP = 210</td><td>TN = 840</td></tr></table> ----- a) 0.8654 b) 0.9221 c) 0.8738 d) 0.8879		Predicted Positive	Predicted Negative	Actual Positive	TP = 950	FN = 120	Actual Negative	FP = 210	TN = 840	2 points
	Predicted Positive	Predicted Negative								
Actual Positive	TP = 950	FN = 120								
Actual Negative	FP = 210	TN = 840								

Q25: You can generate hundreds of trees in a single Random Forest model (denoted as $T_1, T_2, \dots, T_n$). You use a RandomForestClassifier from scikit-learn with the default parameter settings (bootstrap=True, max_samples=None, max_features='sqrt'). Which of the following statements are true? Note: In machine learning, the term "features" is synonymous with "variables" in statistics.

2 points

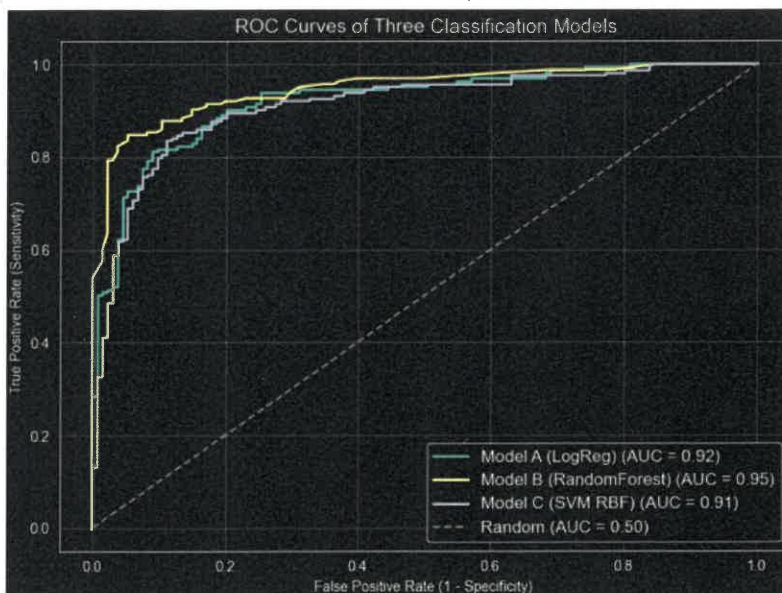
Statements:

- (1) "An individual tree is built on 80% of all features following the 80:20 rule."
- (2) "An individual tree is built on a randomly selected subset of observations."
- (3) "An individual tree considers only a subset of features when looking for the best split."
- (4) "An individual tree is built on 80% of all observations following the 80:20 rule."

- a) Only the statements (1) and (2) are true.
- b) Only the statements (1) and (4) are true.
- c) Only the statements (2) and (3) are true.
- d) Only the statements (3) and (4) are true.

Q26: The following plot shows the ROC curves of three classification models (Model A, Model B, and Model C) evaluated on the same dataset. Which of the following statements is not correct?

2 points



- a) A model with a lower AUC necessarily has higher recall than one with a higher AUC.
- b) A model with an AUC = 0.5 performs no better than random guessing.
- c) The closer a ROC curve lies to the upper-left corner, the better the model performs.
- d) A model with AUC = 1.0 classifies all positive and negative samples perfectly.

Cluster Analysis	
Q27: Which of the following statements about the k-means algorithm is correct? a) k-means can automatically determine the optimal number of clusters. b) k-means minimizes the sum of squared distances between points and their assigned cluster centroids. c) k-means classifies a data point based on the most common class among its k closest neighbors. d) k-means can handle non-numeric (categorical) data without preprocessing.	2 points

Q28: Consider Figure 1, which shows a silhouette plot, and Figure 2, which presents the results of silhouette analyses for different numbers of clusters (from $k = 3$ to $k = 6$). Figure 1 corresponds to one of the k-means clustering results shown in Figure 2. Based on the general interpretation of silhouette plots and the specific results shown in Figures 1 and 2, which of the following statements is **not** correct?

2 points

Figure 1: Silhouette Plot

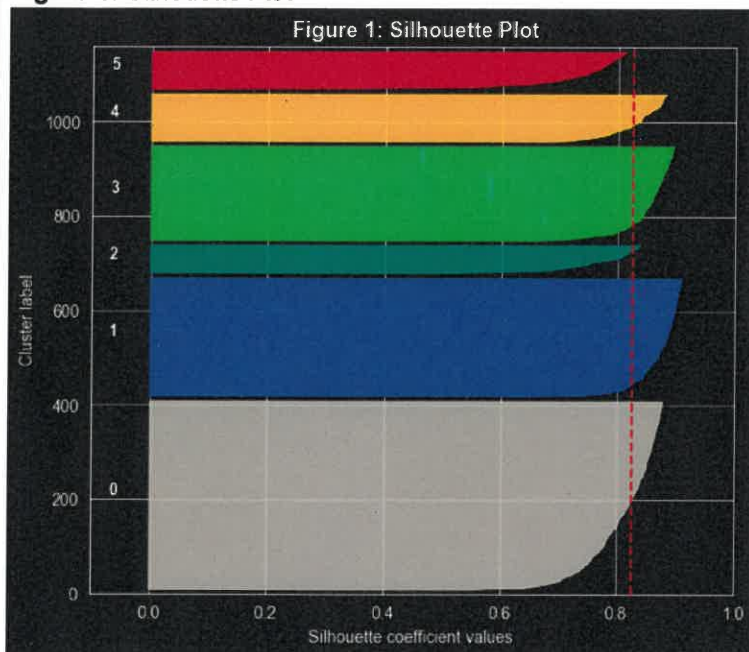


Figure 2: Silhouette Scores for k-means algorithms with $k=3 \dots k=6$:

```
For n_clusters = 3 The average silhouette_score is : 0.5459
For n_clusters = 4 The average silhouette_score is : 0.7052
For n_clusters = 5 The average silhouette_score is : 0.8140
For n_clusters = 6 The average silhouette_score is : 0.8251
```

- a) Figure 1 shows six clusters along with their observations and silhouette coefficient values. The silhouette score is shown as a red dotted line in this figure.
- b) Silhouette coefficients near +1 indicate that the sample is far away from the neighboring clusters, while a coefficient of 0 indicates that the sample is on or very close to the decision boundary between two neighboring clusters.
- c) In Figure 1, the clusters labeled '0' and '1' are the largest clusters as they have the largest number of observations compared to the other clusters.
- d) The k-means clustering algorithm with $k=3$ is the best choice because it provides the lowest silhouette score, indicating that the clusters are better separated than the k-means clustering algorithms with $k=4$, $k=5$, and $k=6$.

Text Mining & Sentiment Analysis																																						
Q29: Consider the sentence, "Lisa bought an expensive car." Which of the following options corresponds to complete preprocessing of the sentence according to the text preprocessing steps presented in class? a) ["lisa", "buy", "an", "expensive", "car"] b) ["lisa", "buy", "expensive", "car"] c) ["Lisa bought an expensive car." d) ["Lisa", "bought", "an", "expensive", "car"]		2 points																																				
Q30: The following table shows the TF-IDF values for five terms (t_1–t_5) across five documents (D1–D5). Based on this matrix, which of the following statements is <u>not</u> correct? TF-IDF matrix <table><tr><th></th><th>t_1 (data)</th><th>t_2 (science)</th><th>t_3 (machine)</th><th>t_4 (learning)</th><th>t_5 (the)</th></tr><tr><td>D1</td><td>0.35</td><td>0.28</td><td>0.00</td><td>0.00</td><td>0.02</td></tr><tr><td>D2</td><td>0.00</td><td>0.38</td><td>0.00</td><td>0.00</td><td>0.03</td></tr><tr><td>D3</td><td>0.42</td><td>0.00</td><td>0.36</td><td>0.29</td><td>0.01</td></tr><tr><td>D4</td><td>0.00</td><td>0.00</td><td>0.41</td><td>0.48</td><td>0.02</td></tr><tr><td>D5</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.01</td></tr></table> a) In document D3, the most important terms are 'data' and 'machine'. b) Document D4 is strongly associated with 'machine' and 'learning'. c) The TF-IDF values of the term 'the' indicate strong importance across documents. d) The term 'science' is more relevant to D2 than to D1.		t_1 (data)	t_2 (science)	t_3 (machine)	t_4 (learning)	t_5 (the)	D1	0.35	0.28	0.00	0.00	0.02	D2	0.00	0.38	0.00	0.00	0.03	D3	0.42	0.00	0.36	0.29	0.01	D4	0.00	0.00	0.41	0.48	0.02	D5	0.00	0.00	0.00	0.00	0.01		2 points
	t_1 (data)	t_2 (science)	t_3 (machine)	t_4 (learning)	t_5 (the)																																	
D1	0.35	0.28	0.00	0.00	0.02																																	
D2	0.00	0.38	0.00	0.00	0.03																																	
D3	0.42	0.00	0.36	0.29	0.01																																	
D4	0.00	0.00	0.41	0.48	0.02																																	
D5	0.00	0.00	0.00	0.00	0.01																																	

END OF EXAM